TASK 2 – BASIC FIREWALL CONFIGURATION WITH UFW

Track: Cybersecurity | Level 1 – Task 2

Objective

Set up and configure the Uncomplicated Firewall (UFW) on Kali Linux, apply rules that allow or deny selected network traffic, and verify that the firewall is enabled. The work was performed in a Kali Linux virtual machine using the terminal.

Tech Stack / Tools

Kali Linux • UFW • Linux Terminal • Oracle VirtualBox

1. Install UFW and Verify Installation

UFW was installed using the package manager after updating the package lists. The version command confirmed that UFW 0.36.2 was installed.

Commands used

```
sudo apt update
sudo apt install ufw -y
sudo ufw version
```

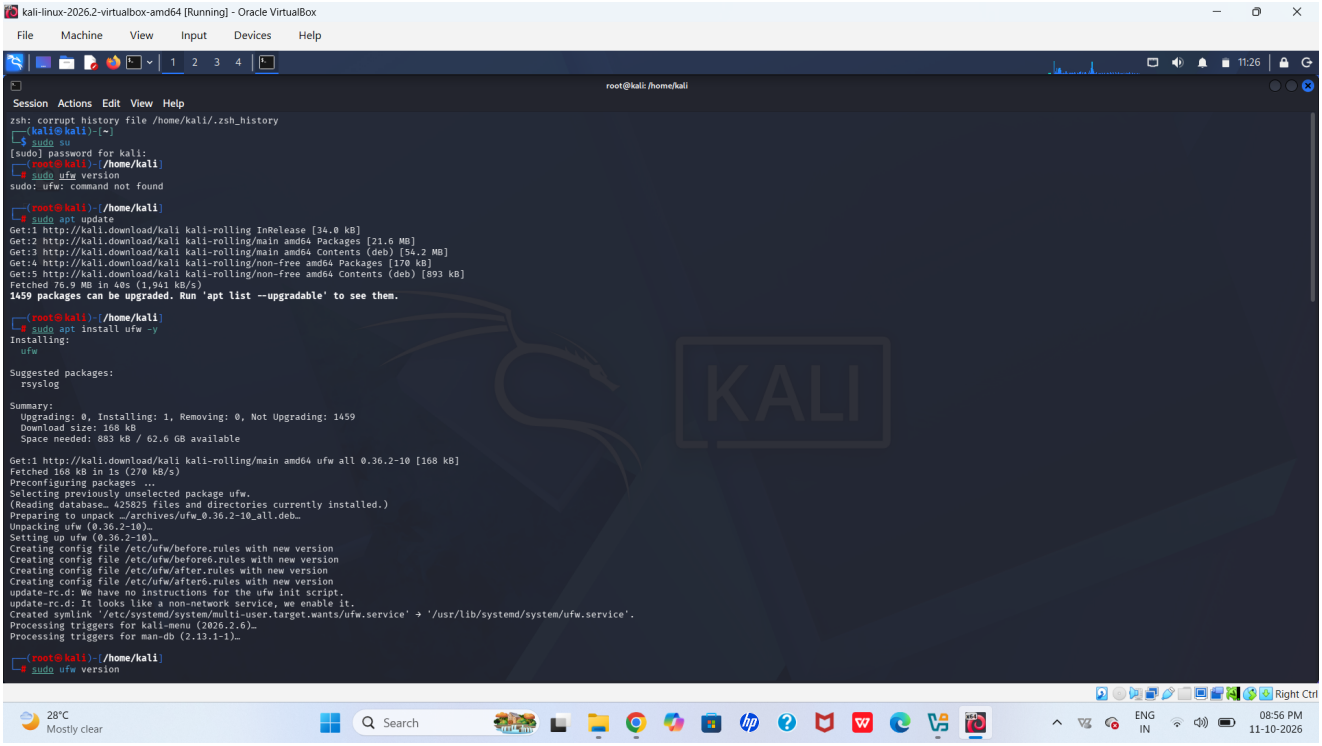


Figure 1. UFW installation and version verification in Kali Linux.

2. Configure Firewall Rules

The default incoming policy was set to deny and the outgoing policy to allow. SSH was allowed, HTTP was denied, HTTPS was allowed, and an additional deny rule was added for the documentation-only example IP address 192.0.2.10.

Command / Rule	Purpose
Default: deny incoming	Blocks incoming connections unless a rule explicitly permits them.
Default: allow outgoing	Allows outbound connections initiated from this system.
allow ssh (TCP 22)	Permits SSH remote administration. Only keep this open where remote access is needed.
deny http (TCP 80)	Denies incoming HTTP traffic on port 80.
allow 443/tcp	Permits incoming HTTPS traffic on port 443.
deny from 192.0.2.10	Adds a source-address deny rule. 192.0.2.10 is a documentation/example address, not a real attacker IP.

Commands used

```
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow ssh
sudo ufw deny http
sudo ufw allow 443/tcp
sudo ufw deny from 192.0.2.10
sudo ufw enable
sudo ufw status verbose
```

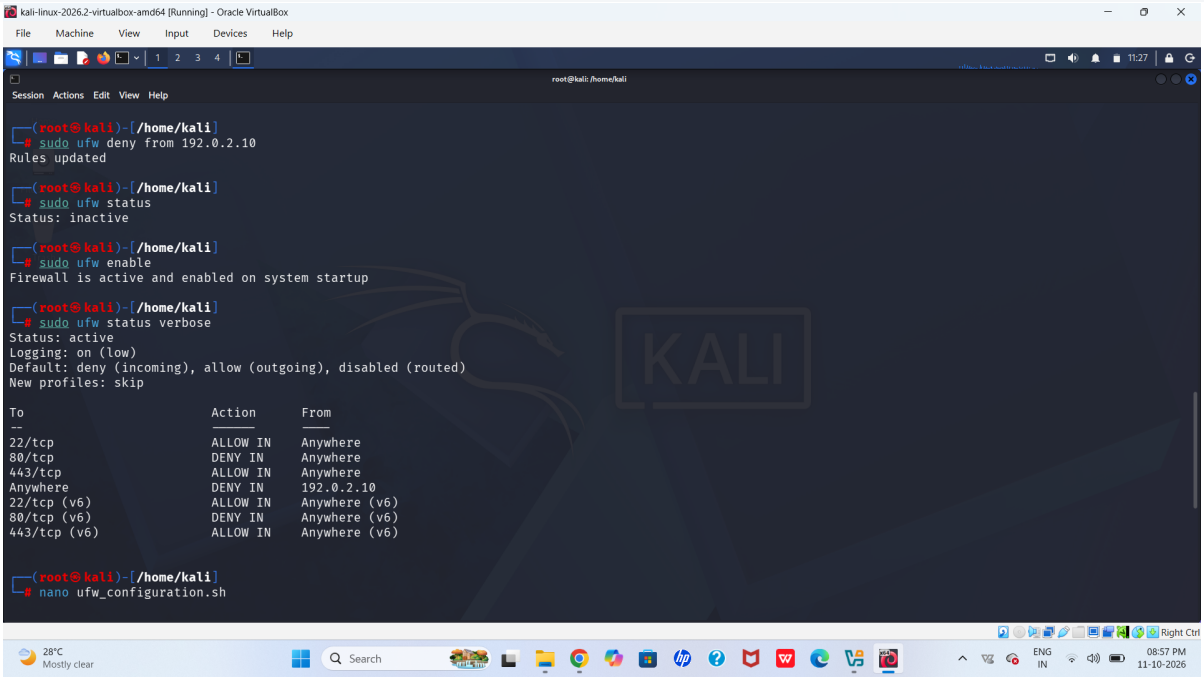


Figure 2. UFW status shows the firewall is active and lists the configured rules.

3. Firewall Rule Configuration Evidence

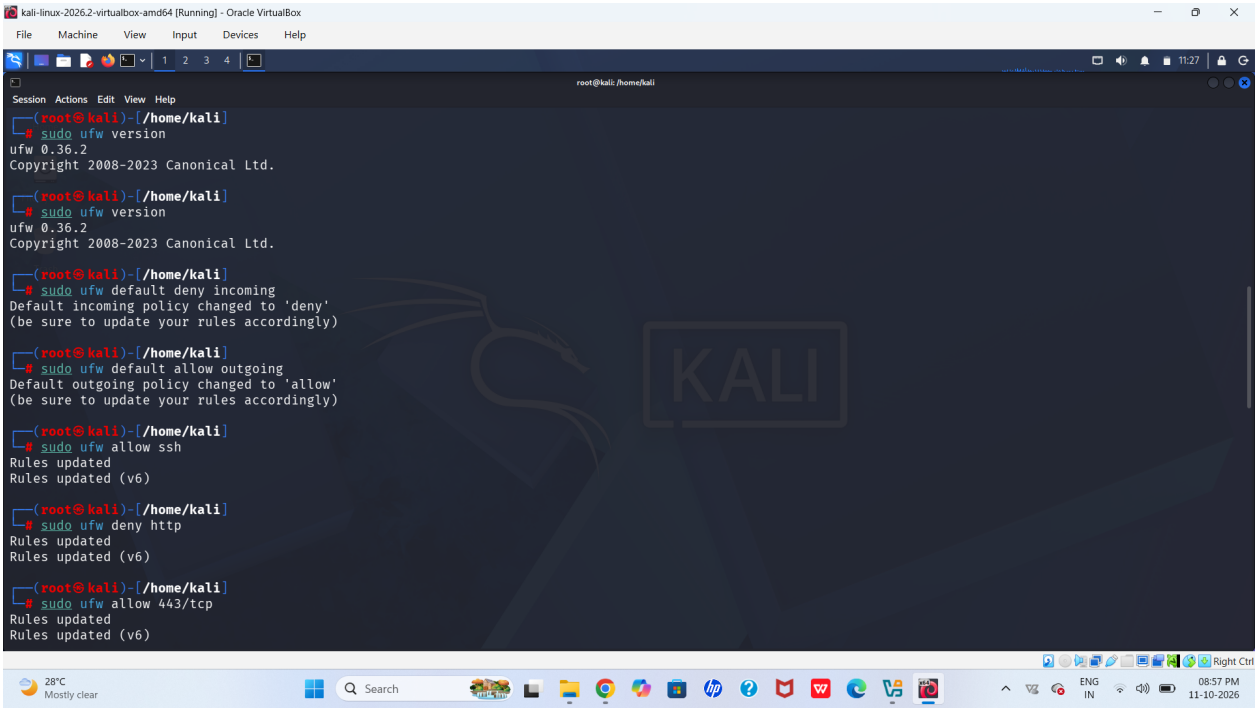


Figure 3. Terminal output showing default policies and SSH, HTTP, and HTTPS rules being added.

4. Automation Script: ufw_configuration.sh

The task requires a runnable script that applies the rules in sequence. The following is the intended script content; save it as **ufw_configuration.sh** and test it before submitting the script file.

```
#!/bin/bash
set -e
if [ "$EUID" -ne 0 ]; then
echo "Please run this script using sudo."
exit 1
fi
ufw default deny incoming
ufw default allow outgoing
ufw allow ssh
ufw deny http
ufw allow 443/tcp
ufw deny from 192.0.2.10
ufw --force enable
ufw status verbose
```

5. Verification and HTTP Blocking Test

The screenshot confirms **Status: active**, the default policies, and the displayed allow/deny rules. However, the screenshots provided do not show a separate HTTP connection test, so they do not by themselves prove that denied HTTP traffic was blocked end-to-end. Before claiming a successful traffic test, run a controlled test against a test web server on a separate authorized VM/device, or clearly document that an external HTTP-blocking test was not completed.

6. Conclusion

UFW was installed and enabled on Kali Linux. The configured policy denies incoming traffic by default and allows outgoing traffic, with explicit rules for SSH, HTTP, HTTPS, and one example source IP. These rules demonstrate basic firewall administration. In a real deployment, allow only the services required, restrict SSH to trusted sources where possible, and verify filtering using a separate authorized test host.

Reference

Ubuntu Community Help Wiki, “UncomplicatedFirewall (UFW)”: https://help.ubuntu.com/community/UFW